

Updated Day-to-Day Technical Milestone Plan for IROS 2026 submission

01/27/2026

1. Prepare the Technical Roadmap
2. Finalize the Dynamics Model and Assumptions
3. Outline the Core Theorems for the Manuscript.

01/28/2026

1. Draft a Formal Problem Statement: Introduce the system model, assumptions, and key mathematical formulations.
2. Literature review: Cover CBF invariance and robust CBFs; CLF–CBF-QP feasibility; distributed critical-edge detection; and MST/tree-based constraint reduction.
3. Begin the Global Connectivity Theorem Draft: Formulate a theorem guaranteeing global connectivity via maintenance of a connected graph (and state the required conditions/invariants).

01/29/2026

1. Read Literature for Proof Foundations: robust CBFs under disturbances; QP continuity; and connectivity/consensus guarantees.
2. Write proofs: complete the graph-theoretic proofs first, then begin the CBF invariance proof.

01/30/2026

1. Integrate Proofs into a Compact Mathematical Framework
2. Finalize the Simulation Plan and Figure Storyboard: define plots, metrics, and baseline methods.

01/31/2026 - 02/01/2026

1. Update and Consolidate Assumptions, Reduce Proof Length, Produce a Clean Notation Table
2. Expand the Related Work from the ACC paper.
3. Write a plan (what to compare and why) for the simulation plots.

02/02/2026

1. Update the dynamic model; verify QP feasibility on small cases.
2. Continue with the simulation result development.

02/03/2026

1. Complete the full set of experiments and gather the plots and data.
2. Start drafting results text (bullet form) tied to each reviewer's concern

02/04/2026

1. Integrate revised model/problem statement, proofs, and new results into the paper
2. Create a 'response-to-review' checklist to ensure every comment is addressed in the paper.

02/05/2026 - 02/06/2026

1. Improve clarity; tighten proofs; ensure Lyapunov/CLF usage is explicit; define $d_{\min}$ and all parameters.
2. Finalize plots if needed.